Fiscal Year:	FY 2007	**Task Last Updated:**	FY 08/04/2008
PI Name:	Barmatz, Martin Ph.D.		
Project Title:	Microgravity Test of Universality and Scaling Predictions Near the Liquid-Gas Critical Point of 3He		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Low temperature condensed matter physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Pasadena	**State:**	CA
Zip Code:	91109	**Congressional District:**	29
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	OTHER
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No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JPL
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Received NCE to 09/30/2007 per D. Strayer (1/07)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Rudnick, Joseph (University of California) Hahn, Inseob (Jet Propulsion Laboratory) Ulf, Israelsson (Jet Propulsion Laboratory) Zappoli, Bernard (Toulouse Space Centre)		
Grant/Contract No.:	None		
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Performance Goal Text:	To explore the frontiers of physics and unlock mysteries of the universe obscured by Earth's gravity		

Task Description:	Scaling theories describing the behavior of thermophysical properties near critical points were developed using the homogeneity postulate. These models led to the definition of universality classes where critical points of the same class are predicted to have the same critical exponents. Efforts to test these models near a liquid-gas critical point in ground-based laboratories are limited due to the gravity-induced vertical density gradient associated with the divergence of the isothermal compressibility. This density gradient becomes appreciable as the critical point is approached, leading to a significant smearing of the transition. Calculations have shown that in a microgravity environment accurate, gravity free, heat capacity and isothermal susceptibility measurements could be obtained two orders of magnitude in reduced temperature closer to the critical point. Ground-based high-precision measurements performed in the gravity-affected as well as the gravity-free regions can also be used to test recent predictions of crossover equation-of-state models. High-resolution sensors are required for the temperature, pressure, and density measurements. The objectives of this task are to test the scaling equation-of-state predictions at the liquid-gas critical point of 3He. The task objectives will include: 1) Measurements of the heat capacity at constant volume along the critical isochore above and below the transition, and measurements of the isothermal susceptibility along the critical isochore above the transition and along the coexistence curve below the transition to determine the critical amplitudes and exponents associated with these diverging thermodynamic quantities. 2) Measurements of the density dependence of the pressure along the critical isotherm to determine the asymptotic critical amplitude and exponent for this diverging quantity. 3) Measurements along the coexistence curve to determine the asymptotic critical amplitudes and exponents associated with the isothermal susceptibility and the shape of the coexistence curve. 4) A rigorous test of equation-of-state predictions both in the asymptotic and crossover regions. The ability to perform these simultaneous measurements should permit a more accurate determination of the asymptotic critical exponents and amplitudes and test asymptotic equation-of-state predictions. Furthermore, knowledge of the asymptotic amplitudes (and exponents) will permit more stringent tests of crossover theories that apply farther away from the critical point. Critical phenomena exist not only in liquid-gas systems, but also in many other real-world systems such as magnetic systems, binary alloys, superconductors and superfluids. In fact the same scaling theory, being tested by MISTE, could apply to many seemingly different physical systems. Thus a high-precision examination of the scaling theory from MISTE constitutes a fundamental test of our basic understanding of material behavior.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The unambiguous validation (or modification) of asymptotic and crossover equations-of-state models that apply to all simple fluids will provide a major advancement in the field of condensed matter physics.
Task Progress:	Several modifications were made to the MISTE low temperature probe. The existing pressure sensor was repositioned within the experimental cell to minimize gravity effects. We redesigned and fabricated a new type of miniaturized high-resolution thermometer, using the metallized paramagnetic compound of PdMn. This new type of HRT has fast equilibration and is less susceptible to moisture contamination than an original GdCl_3 salt type. We also redesigned and fabricated a new capacitive density sensor using copper coated sapphire disks. The new density sensor has much improved parallelism between two electrodes. It also conducts away any electrical field induced heat within the capacitor gap much faster than previous designs. This upgraded probe was put through elaborate calibration tests in preparation for thermodynamic measurements throughout the critical region. A new probe was also fabricated to provide better thermal isolation in future planned experiments. We have measured the heat capacity along 19 isochores over the density (Rho) range $0.9 < Rho/Rho_c < 1.1$. For each isochore, data were taken using the pulse method in the reduced temperature range $\|T/T_c - 1\| < 0.05$ with 8 to 10 points per decade. We started the measurements from the lowest temperature. After equilibrium was reached at each chosen temperature, a short heat pulse of duration 3 to 4 seconds was sent to the cell to create a small temperature change. We recorded the cooling process for a period of time whose length depended on the thermal relaxation time of the fluid at that temperature. From the temperature versus time associated with the cooling of the cell we could estimate the final new equilibrium cell temperature. The total heat capacity was determined knowing the pulse heat input and resultant cell temperature change. The heat capacity of the fluid was then obtained by subtracting the empty cell heat capacity from measured total heat capacity. The measurements on the 19 isochores were obtained during two different cooling runs. In the first run, the cell valve was only weakly linked to the cell via the filling capillary and during the measurement its temperature was always kept 0.2 K higher than that of the cell. In the second run, we thermally coupled the valve to the cell with copper shim stock and shorten the length of the filling capillary so that the valve would be at the same temperature as the cell. This change was made to reduce some irregular fluid motions observed between the valve and cell during the first run. These measurements throughout the critical region are now being used to test the predictions of the Crossover Parametric Model (CPM) and a field-theoretical Parametric Crossover Model (PCM) equations of state. The results of these analyses will be published. We have also measured the linear susceptibility along the critical isochore and isotherms throughout the critical region of 3He using the electrostriction method. This electrostriction technique uses a dc electric field across a parallel plate capacitor to create an equivalent pressure difference between inside and outside of the capacitor. The linear susceptibility along an isotherm is determined from the measured density change associated with a pressure change induced by dc field. The electrostriction results agree with linear susceptibility measurements using a conventional PVT technique along isotherms and along the critical isochore in the region where the gravity effect is negligible. For isotherms near the critical isotherm, the sharp divergence of the linear susceptibility is rounded due to gravity-induced density stratification. These measurements were successfully interpreted using the crossover parametric model and will be published. Considerable amount of time and effort went into publishing a paper titled Critical Phenomena in Microgravity: Past, Present, and Future, M. Barmatz, J. Lipa, R.V. Duncan, and I. Hahn, Rev. Mod. Phys. 79, 1, (2007). This was a review of the status of critical phenomena studies that included the MISTE work as well as the results of many other researchers throughout the world.
Bibliography Type:	Description: (Last Updated: 08/04/2008)

Articles in Peer-reviewed Journals	Barmatz M, Hahn I, Lipa J, Duncan RV. "Critical phenomena in microgravity: past, present, and future." Reviews of Modern Physics. 2007 Jan-Mar;79(1):1-52. http://dx.doi.org/10.1103/RevModPhys.79.1 , Jan-2007